

SEMESTER III.

MODULE 5

HAMILTONIAN AND

EULERIAN GRAPH

Walk

An alternating sequence of vertices and edges in a graph is called a walk.

In a walk the vertices and the edges may be repeated or may not be repeated.

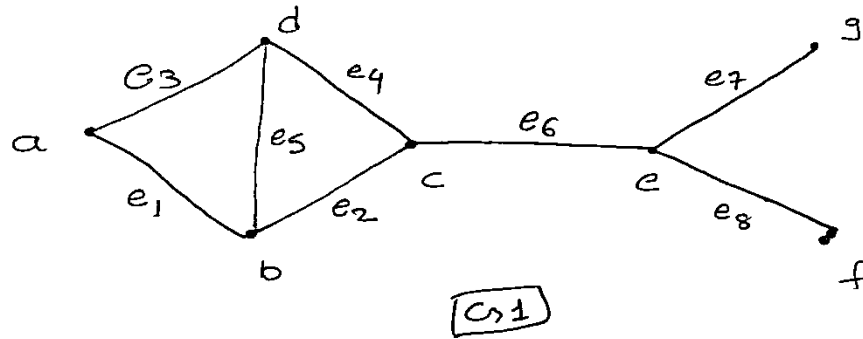
The starting vertex and the end vertex may be or may not be the same.

Thus the walk is the most general concept.

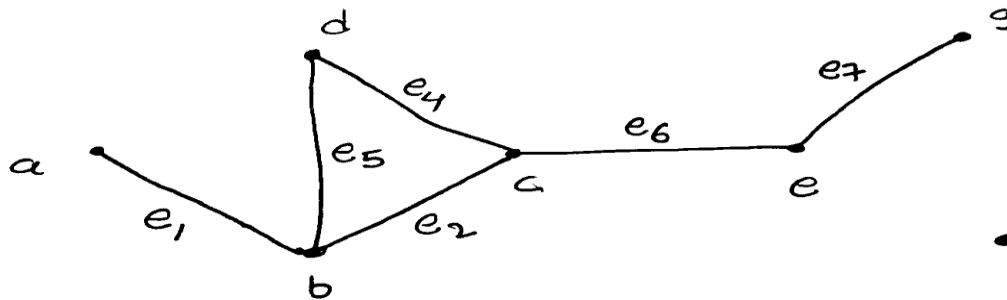
The number of edges in the walk is called the **length** of the walk.

Walk

For example, consider the following graph G_1 ,



In G_1 , $a e_1 b e_2 c e_4 d e_5 b e_2 c e_6 e e_7 g$ is a walk.



Some vertices as well as some edges are repeated.
We can write some more walks according to our will

Trail

If in a walk no edge is repeated then it is called a trail.

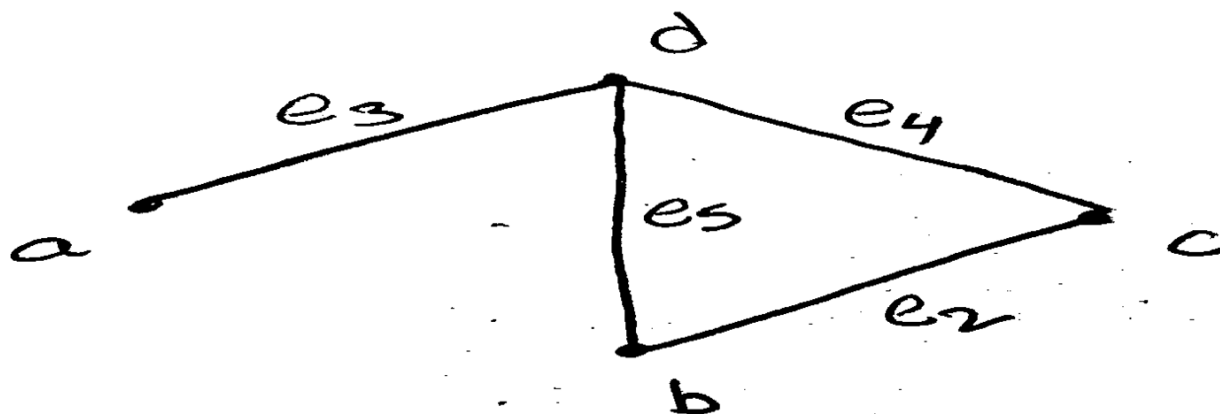
Thus, in a trail vertices may be repeated but edges are not repeated.

For example in G_1 ,

$a e_3 d e_5 b e_2 c e_4 d$ is a trail.

In this the vertex d is repeated but no edge is repeated.

The trail is shown in the following figure

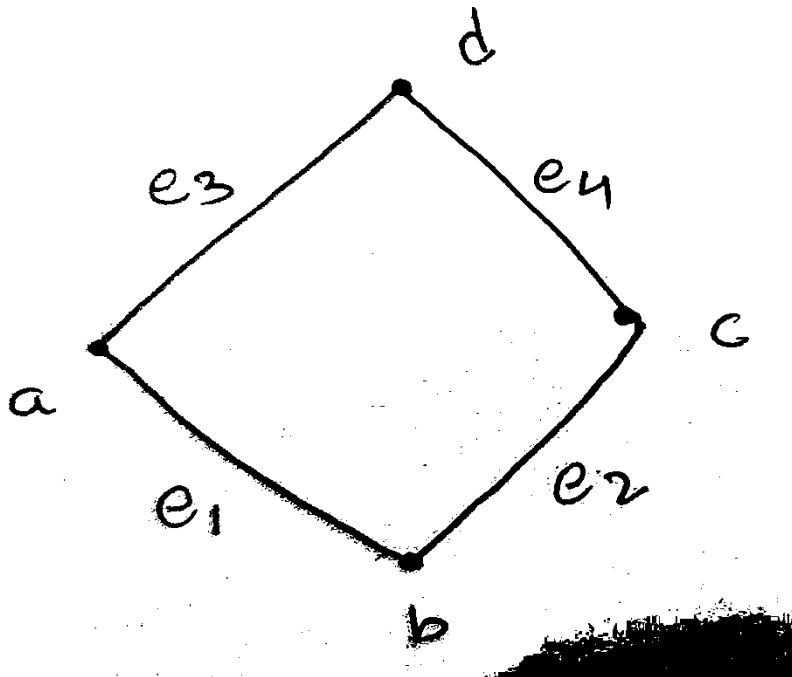


Circuit

A trail in which starting & ending vertex is same is called as circuit.

OR

A walk in which no edge is repeated but vertices can be repeated with same starting & ending vertex is called as circuit.



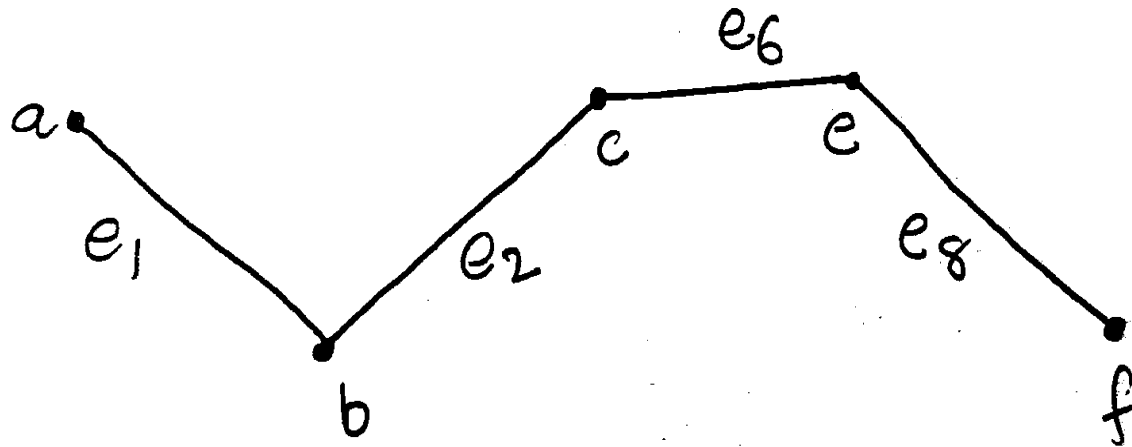
Path

A walk in which neither a vertex nor an edge is repeated is called a Path.

In the graph G_1 ,

$a e_1 b e_2 c e_6 e e_8 f$

Is a path as no vertex is repeated and no edge is repeated .



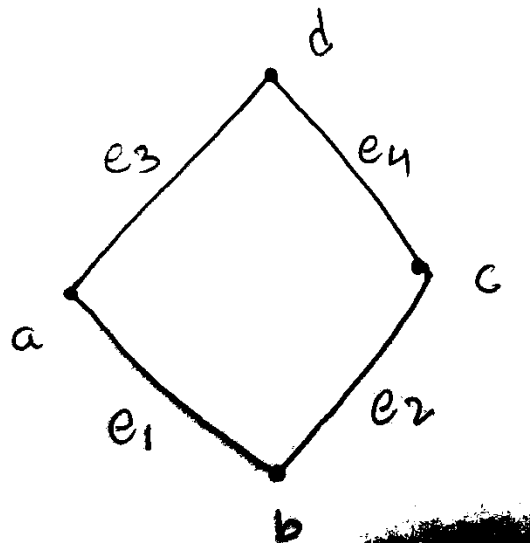
Cycle

A path in which starting & ending vertex is same is called as cycle.

OR

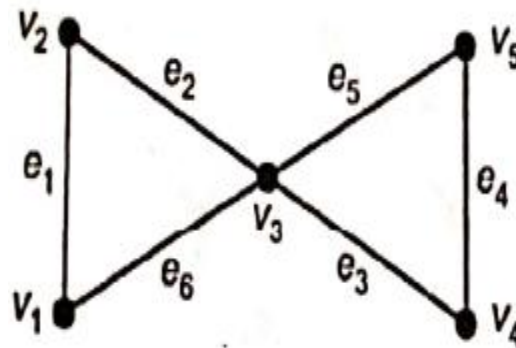
A walk in which no edge is repeated & no vertex is repeated except starting & ending vertex is called as cycle.

In the graph G_1 , $a e_1 b e_2 c e_4 d e_3 a$ is a cycle.



A walk in which no edge is repeated & no vertex is repeated except starting & ending vertex is called as cycle

A walk in which no edge is repeated but vertices can be repeated with same starting & ending vertex is called as circuit.



Example :

$v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_3 e_6 v_1$ is a circuit (not cycle)

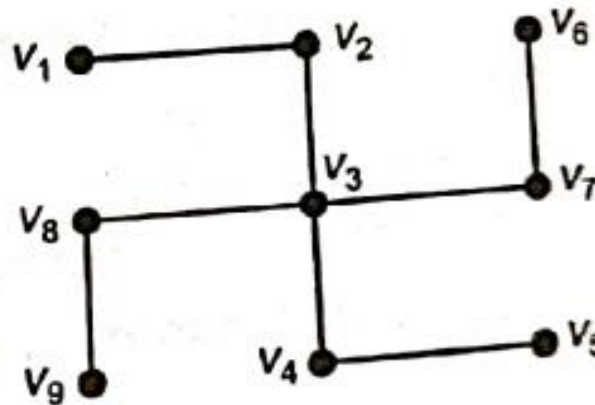
$v_1 e_1 v_2 e_2 v_3 e_6 v_1$ is a cycle (or circuit)

	Repeated Vertex	Repeated Edge
Walk	Allowed	Allowed
Trail	Allowed	No
Path	No	No
Circuit	Allowed	No
Cycle	No (Except Terminal Vertices)	No

Definition : Path

path : A sequence of vertices $[v_0, v_1, v_2, \dots, v_l]$ is a *path* from v_0 to v_l of length l in G if $v_{i-1}v_i \in E$ for $i=1, 2, \dots, l$.

Example



$v_1 v_2 v_3 v_4 v_5$ is a path of length 4

$v_1 v_2 v_3 v_7$ is a path of length 3

Connected graph

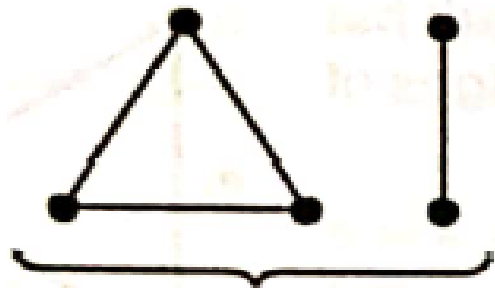
A graph is called connected graph if there is a path from any vertex to any other vertex

Example :



Edge Connectivity

Vertex Connectivity

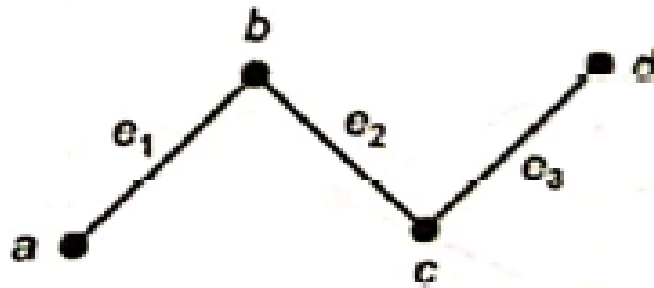


Applications in
Military, traffic and
communications??

Eulerian Path/ Circuit/ Graph

An **Eulerian path** is a path through the graph in which **every** edge of a graph G appears in the path and it appears **exactly once** in a path.

Example :

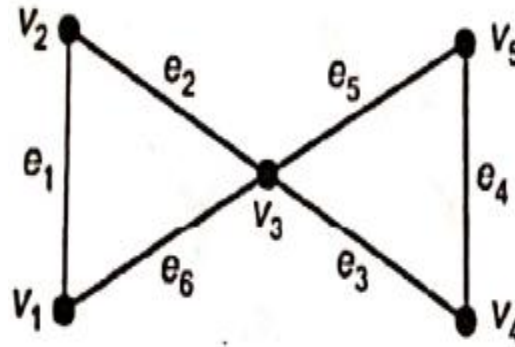


$ae_1 be_2c e_3d$ is an Eulerian path

A circuit in which **every** edge of a graph G appears **exactly once** is called an **Eulerian circuit**.

A Graph which has an Eulerian circuit is called an **Eulerian Graph**.

Example :

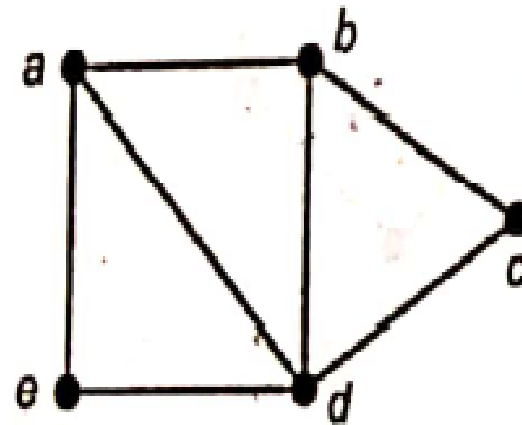
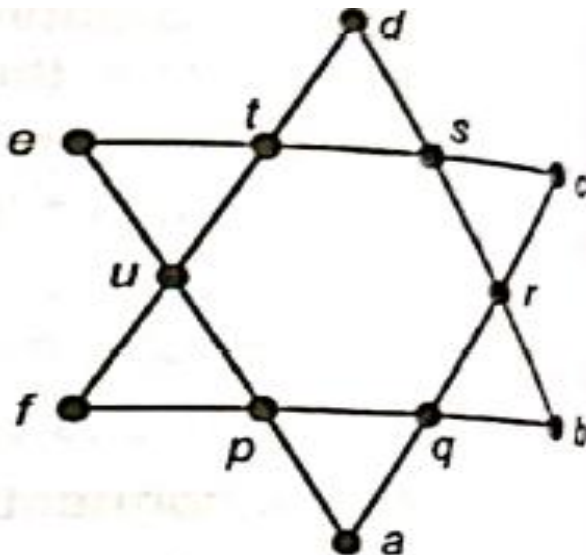


$v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_3 e_6 v_1$ is an Eulerian circuit.

Note, There is No Eulerian cycle in the Graph due to V_3

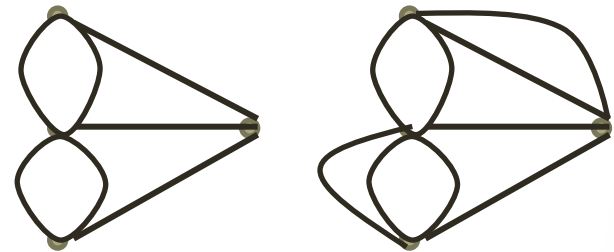
Theorem 1 : G is an Eulerian graph if and only if all vertices of G are of even degree

Theorem 2 : G is a connected graph having exactly 2 vertices (say u, v) of odd degree then G contains Eulerian trail from u to v .



Example : Is Königsberg seven bridge problem solvable or not? What is the minimum number of bridges to be built in order to make it solvable. (Find all possible solutions for them)

ANS. Königsberg problem is isomorphic to following graph. The problem is solvable if we could find an Eulerian circuit in this graph.



But this graph is not Eulerian.
(Check the degrees of vertices)

Therefore Seven bridge problem is not solvable.

We can make it solvable by making degrees of each vertex even. (One solution is depicted. How many different ways are possible?)

Hamiltonian Graph

Let $G = (V, E)$ be a graph.

A path in G which contains all the vertices of G is called a **Hamiltonian Path**.

A circuit in G which contains all the vertices in G is called a **Hamiltonian circuit**.

If G has a Hamiltonian path then G is called a **semi-Hamiltonian Graph**.

If G has a Hamiltonian cycle, then G is called a **Hamiltonian Graph**.

Hamiltonian Graph

Notes:

1. Every Hamiltonian graph clearly is a semi Hamiltonian Graph.
2. A semi Hamiltonian graph need not be Hamiltonian
3. Like an Eulerian graph, a Hamiltonian (and Semi Hamiltonian) graph is connected.

Hamiltonian Graph

Notes:

4. There is a difference between an Eulerian circuit and a Hamiltonian circuit.

In Eulerian circuit each edge must be traversed.

Hence, in an Eulerian circuit each vertex will be visited but some vertices might be visited more than once.

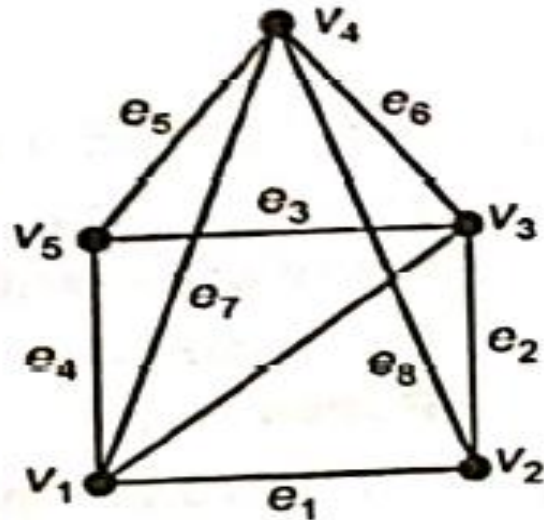
But in Hamiltonian circuit all the vertices will be visited once and once only except the terminal vertex.

Hence, some edges may not be traversed at all

5. We have a very simple test to find out whether a graph is Eulerian viz. each vertex has even degree.
6. But there is no such simple rule to decide whether a graph is Hamiltonian.

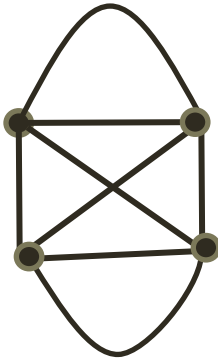
	Repeated Vertex	Repeated Edge	Edges	Vertices
Eulerian	Allowed	No	All	All
Hamiltonian	No (Except Terminal)	Allowed	Not necessarily all	All

Example :

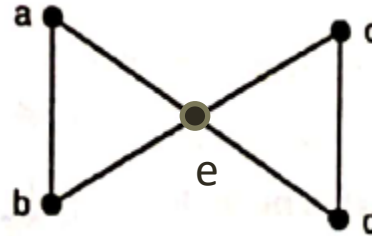


$v_1 e_1 v_2 e_2 v_3 e_6 v_4 e_5 v_5 e_4 v_1$ is Hamiltonian cycle

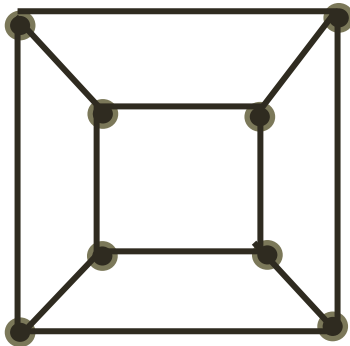
Examples: Check for Eulerian and Hamiltonian



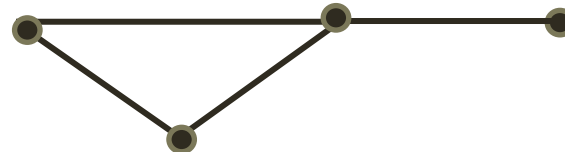
E and H



E but NH



NE but H



NE and NH

Theorem 2 : If G is a simple connected graph with n vertices, where $\deg(x) \geq (n/2)$ for each vertex, then the graph G is Hamiltonian graph.

Example: K_5 , check previous examples

Note : converse is not true.



Travelling Salesperson Problem (T.S.P.)

One of the applications of Hamiltonian cycle is Travelling Salesperson Problem (T.S.P.)

A salesperson is required to start from the headquarter V and visit the retail shops located at a,b,c,d,..... then come back to the headquarter at V . If the **distance** from any place to any other place is **known**, then what should be the **order of the places** in which salesperson should visit the places so that **distance travelled is minimum**. This is called as Travelling Salesperson Problem

Travelling Salesperson Problem is solved by the method called as nearest neighbour method, procedure is as follows.

1. Start from vertex V which denotes the headquarter.

From V go to the vertex which has minimum distance from V .

2. If V_1 is such vertex then join V to V_1 by an edge.

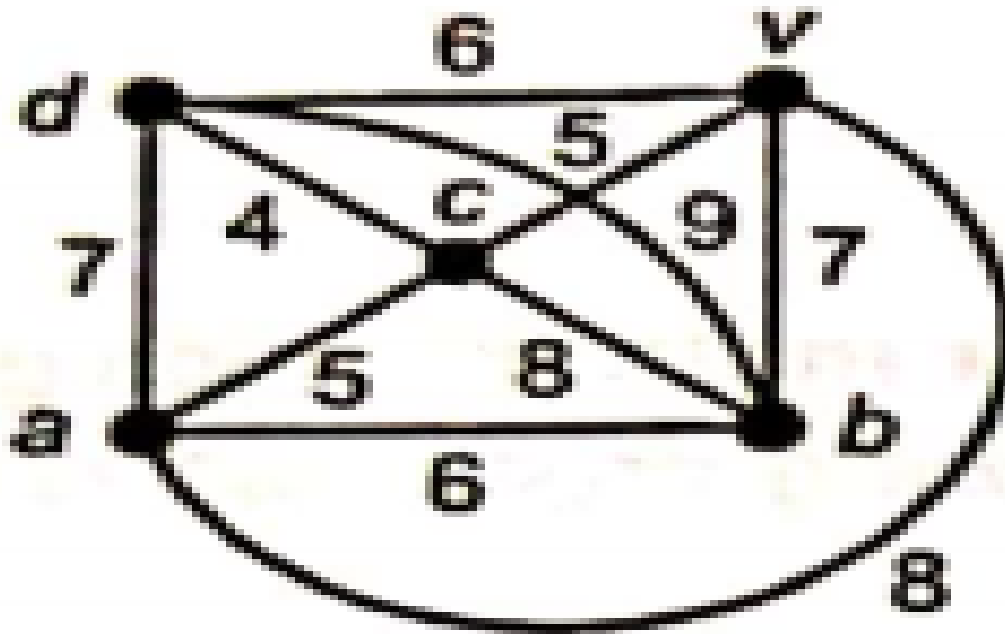
From V_1 go to the vertex which has minimum distance from V_1 .

3. If V_2 is such vertex then join V_1 to V_2 by an edge.

4. In this way complete Hamiltonian cycle, joining the last vertex to V back.

Example :

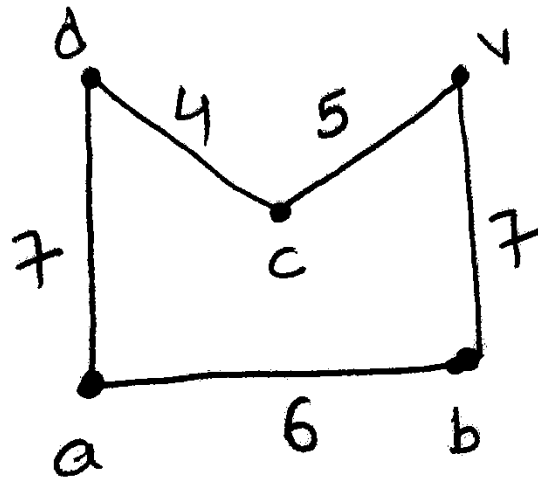
Solve Travelling salesperson problem for the given graph with starting vertex as V and hence find the length of a required Hamiltonian cycle.



Answer :

1. Vertex V is adjacent with all vertices a, b, c, d , but distance between V & c is minimum. To construct Hamiltonian cycle, consider first edge Vc .
2. Vertex c is adjacent with vertices a, b, d but distance between c & d is minimum. To construct Hamiltonian cycle, consider second edge cd .
3. Vertex d is adjacent with vertices a, b , but distance between d & a is minimum. To construct Hamiltonian cycle, consider third edge da .
4. Vertex a is adjacent with remaining vertex b . To construct Hamiltonian cycle, consider next edge ab .
5. To complete Hamiltonian cycle, joining the last vertex b to V .

We get Hamiltonian cycle as follows



The length of a required Hamiltonian cycle V-c-d-a-b-V is $5+4+7+6+7=29$ units

Example :

Find the minimum Hamiltonian cycle for the given graph with starting vertex (i) as a and (ii) as d hence find the length of a required Hamiltonian cycle.

